Superposition in Circuit Analysis

Trieu Truong

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I. Motivation

During my second year, the Superposition Priciple in circuit analysis always fascinated me because it can be used to solve both linear and non-linear problems by considering the effect of each independent source separately. Therefore, I am writing this paper with a belief that it could be taught in a more systematic approach with more advanced problems.

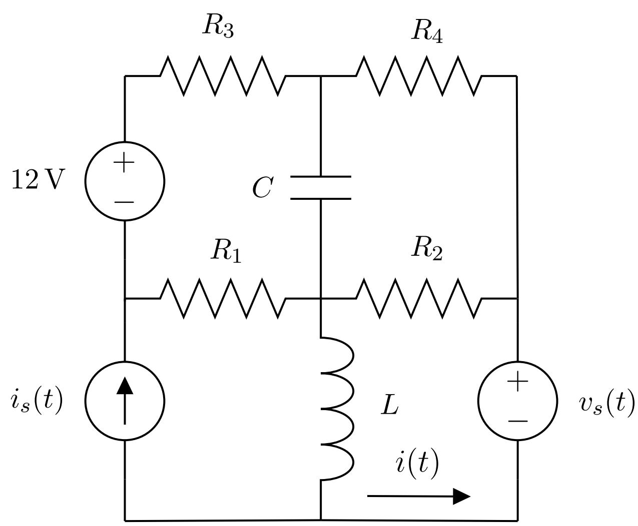
II. Introduction

You might have known about the Superposition Principle since it appears in many forms of mechanics or electromagnetics. In circuit analysis, the principle states

In a linear circuit containing multiple independent sources, the voltage or current across any circuit element is equal to the algebraic sum of the responses caused by each independent source acting individually.

III. Applications

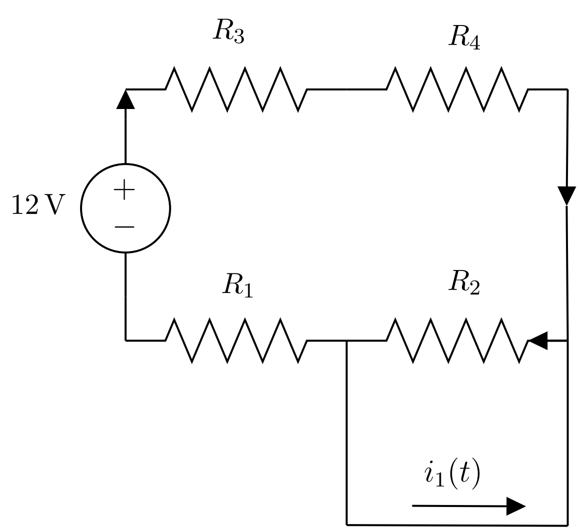
Given a circuit with the source $i_s(t) = 4\cos(3t)\,A$ and $v_s(t) = 3\sin(4t)\,V$, where the circuit parameters are $R_1 = R_2 = R_3 = R_4 = 1\,\Omega$, $L = 1H$ and $C = \dfrac{1}{12}\,F$. Use superposition theorem to solve for the current $i(t)$.



Solution. We first solve for DC signal. In DC steady state,

- We turn off other **current sources** so they become **open circuits** and turn off other **voltage sources** so they become **short circuits**
- Capacitor acts like an **open circuit**, inductor acts like a **short circuit**

Thus, the diagram is simplified as

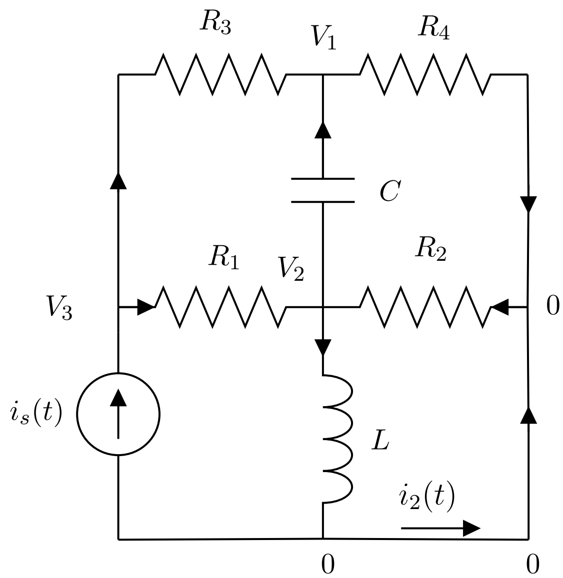


We can easily compute current $i(t)$ in DC steady state by using Ohm's law

$$i_1(t) = -\frac{12}{R_1 + R_3 + R_4} = -\frac{12}{1+1+1} = -4\,\mathrm{A}$$

Noted that the minus sign indicates the current in this state is flowing in the opposite direction of the given problem.

Next, we solve the current in AC steady state due to the current source. The same principle applies here, we turn off other sources



According to Kirchhoff's Current Law,

$$\frac{V_2 - V_1}{1/(i\omega C)} + \frac{V_3 - V_1}{R_3} = \frac{V_1 - 0}{R_4} \tag{1}$$

$$\Rightarrow \frac{i}{4}(V_2 - V_1) + V_3 - V_1 = V_1$$

$$\frac{V_3 - V_2}{R_1} + \frac{0 - V_2}{R_2} = \frac{V_2 - V_1}{1/(i\omega C)} + \frac{V_2 - 0}{i\omega L} \tag{2}$$

$$\Rightarrow V_3 - V_2 - V_2 = \frac{i}{4}(V_2 - V_1) - \frac{i}{3}V_2$$

$$\frac{V_3 - V_2}{R_1} + \frac{V_3 - V_1}{R_3} = I_s \tag{3}$$

$$\Rightarrow V_3 - V_2 + V_3 - V_1 = 4\angle 0$$

Solving this system of equations, we yield

$$V_1 = \frac{20-8i}{10-5i}, \qquad V_2 = \frac{48+12i}{25}, \qquad V_3 = \frac{98+8i}{25}.$$

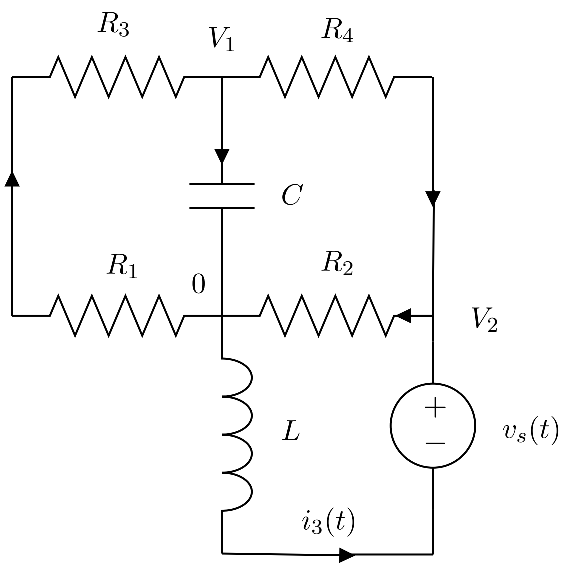
Therefore, we can solve for phasor I_2

$$I_2 = \frac{V_2 - 0}{i\omega L} - I_s = \frac{1}{3i}\frac{48+12i}{25} - 4\angle 0 = -3.84 - 0.64i$$

So we can convert it back to time domain

$$i_2(t) = \frac{16}{25}\sqrt{37}\cos\left(3t + \pi + \arctan\frac{1}{6}\right)$$

Lastly, we focus on the AC voltage source by turning off other sources,



Solving the last piece is just a repetitive work. According to Kirchhoff's Current Law,

$$\frac{0-V_1}{R_1+R_3} = \frac{V_1 - 0}{1/(i\omega C)} + \frac{V_1 - V_2}{R_4} \tag{4}$$

$$\Rightarrow -\frac{V_1}{2} = \frac{i}{3}V_1 + V_1 - V_2$$

$$\frac{V_2 - 0}{R_2} = \frac{V_1 - V_2}{R_4} + \frac{0 + V_s - V_2}{i\omega L} \tag{5}$$

$$\Rightarrow V_2 = V_1 - V_2 + \frac{3\angle 0 - V_2}{4i}$$

Solving this system of equations, we yield

$$V_1 = \frac{-8}{7-50i}, \qquad V_2 = \frac{-27-6i}{7-50i}.$$

Therefore, we solve for phasor I_3

$$I_3 = \frac{0+V_s-V_2}{i\omega L} = \frac{1}{4i}\left(3\angle 0 - \frac{-27-6i}{7-50i}\right) = \frac{1312}{10196} - \frac{10085}{10196}i.$$

So we can convert it back to time domain

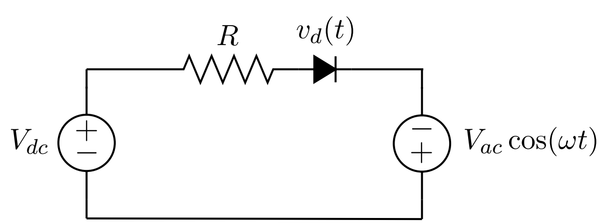
$$i_3(t) = \frac{12\sqrt{10}}{\sqrt{2549}}\sin\left(4t + \arctan\frac{157}{29}\right)$$

By superposition, we just need to add up three signals, then we are done!

$$i(t) = i_1(t) + i_2(t) + i_3(t) \approx -4 + 3.893\cos(3t + 2.976) + 0.752\sin(4t + 1.388)$$

As you can see, the solution is a bit tedious...especially if the problem has more than 4 independent sources since you have to redraw the circuit over again and find the current for each source. In fact, you can solve this problem using Laplace Transform to get the transient and steady state response at the same time (it is encouraged to try on your own as an exercise). So what is the point of learning this tool? The next problem will have an answer to this question.

Problem 1.2. Given the circuit with $V_{ac} = 4$ V, $V_{dc} = 1$ V and $R = 3\,\Omega$. A diode exhibits a voltage of 0.7 V at 1.6 mA with $n = 2$ and $V_T = 0.025$ V. Calculate voltage of the real diode in the form of $v_d(t) = \alpha + \beta\cos(\omega t)$. Note: That there are no switches drawn in the figure signifies that the circuit has been as shown for a long time, that is, it is in steady state.



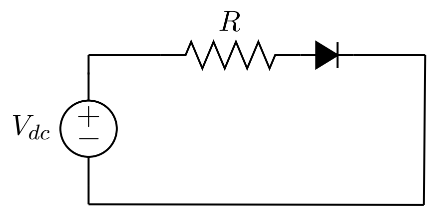
Solution. Before tackling this problem using superposition, I want to draw your attention to the direct approach. According to Kirchhoff's Voltage Law, we can write

$$\begin{aligned} V_{dc} + V_{ac}\cos(\omega t) &= v_d + i_d R \\ i_d &= I_s(e^{v_d/(nV_T)} - 1) \end{aligned}$$

When we write these equations, it may seem that having two equations and two unknowns should be enough to solve the system directly. However, the situation is more complicated. The diode equation is nonlinear because of its exponential dependence on v_d. In general, such systems do not admit simple closed-form solutions and are usually solved using numerical methods or suitable approximations such as small-signal linearization. So we will use superposition with the second approach.

However, you might think: "Superposition is only valid for linear systems." That is true. This is precisely why we linearize the circuit around its operating point. After linearization, the circuit behaves approximately linearly for small AC variations, allowing the DC bias and AC signals to be analyzed separately using superposition.

In DC steady state, we can turn off the AC source. The diagram can be redrawn as



We first find the reverse saturation current I_s

$$I_s \approx \frac{I_d}{e^{V_D/(nV_T)}} = \frac{1.6 \times 10^{-3}}{e^{0.7/(2 \cdot 0.025)}} \approx 1.33 \times 10^{-9}\text{A}$$

Since V_{dc} is significantly large, we apply Kirchhoff's Voltage Law

$$V_{dc} = \alpha + Ri_d$$

Then our goal is to solve the non-linear equation

$$\alpha + 3I_s(e^{\alpha/(2 \cdot 0.025)} - 1) = 1$$

Let $f(\alpha) = 1 + 3I_s(e^{20\alpha} - 1) - 1 \Rightarrow f'(\alpha) = 1 + 60I_s e^{20\alpha}$

We can use Newton-Raphson method to approximate the solution

$$\alpha_{k+1} = \alpha_k - \frac{f(\alpha_k)}{f'(\alpha_k)} \quad \text{with} \quad \alpha_0 = 0.7\,V$$

Then the equation becomes

$$\alpha_{k+1} = \frac{1 + 3I_s(e^{20\alpha_k} - 1) - 1}{1 + 60I_s e^{20\alpha_k}} \quad \text{with} \quad \alpha_0 = 0.7\,\mathrm{V}$$

The first six iterates are

$$\begin{array}{lll} \alpha_1 = 0.9693 & \alpha_2 = 0.9230 & \alpha_3 = 0.8867 \\ \alpha_4 = 0.8692 & \alpha_5 = 0.8664 & \alpha_6 = 0.8663 \end{array}$$

As the number of iterations increases, the sequence converges to

$$\alpha \approx 0.8663\,\mathrm{V}$$

Now we recall the Shockley diode equation

$$i \approx I_s e^{v/(nV_T)} \Rightarrow 1 = \frac{I_s}{nV_T} e^{v/(nV_T)} \frac{\mathrm{d}v}{\mathrm{d}i}$$

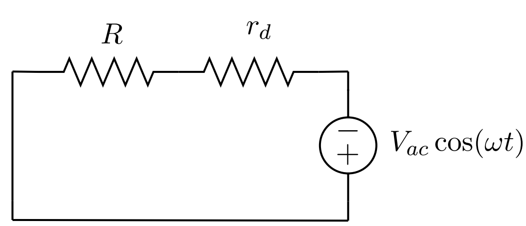
Rearranging two sides of the equation,

$$\frac{\mathrm{d}v}{\mathrm{d}i} = \frac{nV_T}{I_s e^{\alpha/(nV_T)}} = \frac{nV_T}{I_Q}$$

The derivative $\mathrm{d}v/\mathrm{d}i$ is called dynamic resistance of the diode. The subscript Q refers to the quiescent point, which represents the steady-state DC operating condition of an electronic device. Therefore,

$$r_d = \frac{nV_T}{I_Q} = \frac{2 \cdot 0.025}{1.33 \times 10^{-9} \cdot e^{0.866/(2 \cdot 0.025)}} \approx 1.12\,\Omega$$

Now we find the voltage of the diode in AC steady state by replacing the diode with its dynamic resistor



Using voltage divider, we yield

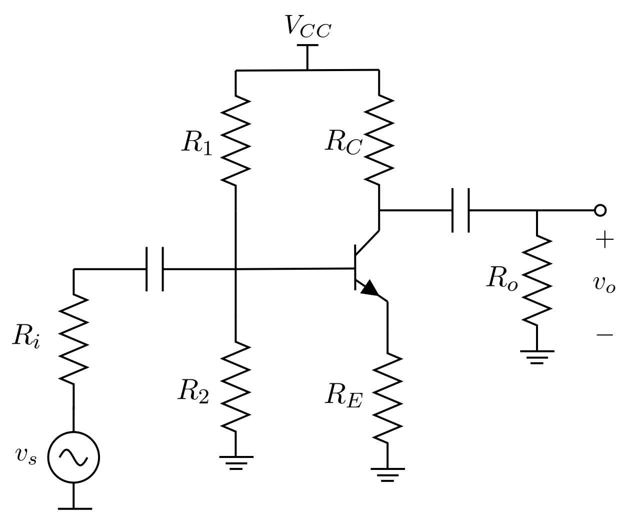
$$\beta = \frac{r_d}{r_d + R} V_{ac} \cos(\omega t) = \frac{1.12}{1.12 + 3} \cdot 4 \cos(\omega t) \approx 1.09 \cos(\omega t)\,\text{V}$$

Therefore, by superposition

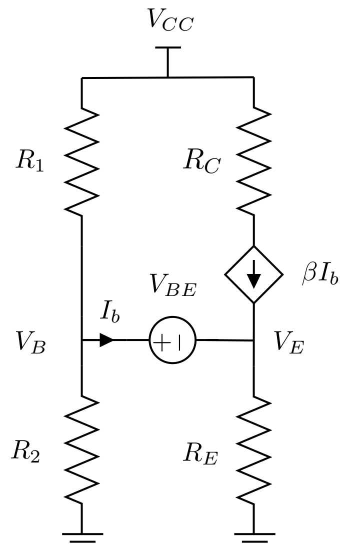
$$v_d(t) = 0.866 + 1.09 \cos(\omega t)\ \text{V}$$

So far, we can safely conclude that the Superposition Principle is extremely useful for analyzing small-signal models of nonlinear devices such as diodes, BJTs, and MOSFETs. However, because these devices are inherently nonlinear, the principle becomes valid only after linearization around an operating point, and therefore applies only as an approximation for sufficiently small signals.

Problem 1.3. Given transistor in an amplifier below with $\beta = 100$, $V_{BE} = 0.7$ V, and $V_{CE(sat)} = 0.2$ V. Meanwhile, $V_{CC} = 5$ V,$R_i = 50\,\Omega, R_o = 110\,\Omega, R_1 = 6\,\text{k}\Omega, R_2 = 4\,\text{k}\Omega, R_C = 800\,\Omega$ and $R_E = 640\,\Omega$. Assume $V_T = 0.025$ V. If the input signal is $v_i(t) = 10\sin(2t)$ determine the output voltage $v_o(t)$



Solution. For DC analysis, we turn off the AC source and the coupling capacitors become open circuits. Therefore, the circuit diagram is simplified to



You might ask why we should not use the more accurate non-linear model $I_C = I_s e^{V_{BE}/V_T}$ for this analysis. While this model is indeed more precise, it leads to much more complicated equations. Therefore, when the BJT operates in the active region, we typically use the simpler 0.7 V approximation for practical DC analysis and this is good enough. According to Kirchhoff's Current Law,

$$\frac{V_{CC} - V_B}{R_1} = I_B + \frac{V_B}{R_2} \tag{6}$$

$$\Rightarrow \frac{5 - V_B}{6000} = I_B + \frac{V_B}{4000}$$

$$(\beta + 1) I_B = \frac{V_E}{R_E} \tag{7}$$

$$\Rightarrow 101 I_B = \frac{V_E}{640}$$

$$V_B - V_E = V_{BE} \tag{8}$$

$$\Rightarrow V_B - V_E = 0.7$$

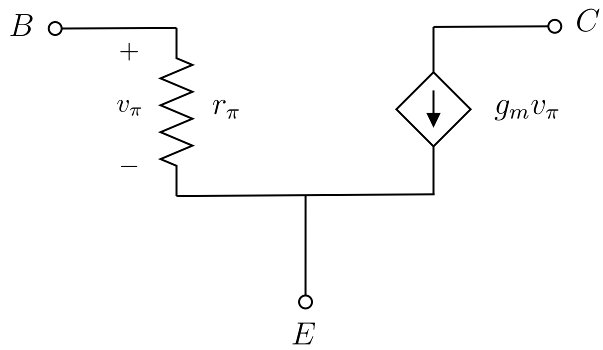
Solving this system of equations, we yield

$$V_B \approx 1.953\,\mathrm{V}, \qquad V_E = 1.253\,\mathrm{V}, \quad I_B = 1.939 \times 10^{-5}\,\mathrm{A}$$

As a result, $V_C = V_{CC} - \beta I_B R_C = 5 - 100 \cdot 1.939 \times 10^{-5} \cdot 800 \approx 3.449$ V.

Since $V_{CE} = V_C - V_E \approx 2.2\,\text{V} > V_{CE(sat)} = 0.2\,\text{V}$, the BJT is in active region and our assumption was correct.

For AC analysis, we short the constant voltage source and coupling capacitors become ideal wires. Additionally, we introduce the hybrid-π model, which is a a small-signal equivalent circuit used to analyze a BJT transistor for AC signals.



Now we will use the non-linear BJT equation to derive two important parameters: the small-signal resistance seen looking into the base-emitter junction r_π and small-signal transconductance g_m.

The small-signal transconductance of a BJT is defined as

$$g_m = \frac{\mathrm{d}I_c}{\mathrm{d}V_{BE}}$$

This derivative can be found from the Shockley diode equation

$$I_C = I_s e^{V_{BE}/V_T}$$

Differentiating two sides of the equation with respect to V_{BE},

$$\frac{\mathrm{d}I_c}{\mathrm{d}V_{BE}} = \frac{I_s}{V_T} e^{V_{BE}/V_T} = \frac{I_{CQ}}{V_T}$$

Therefore,

$$g_m = \frac{I_{CQ}}{V_T}$$

Meanwhile, small-signal resistance seen looking into the base-emitter junction r_π is defined as

$$r_\pi = \frac{\mathrm{d}V_{BE}}{\mathrm{d}I_C}\frac{\mathrm{d}I_C}{\mathrm{d}I_B} = \frac{\beta}{g_m}$$

Since $g_m = I_{CQ}/V_T = \beta I_{BQ}/V_T$, we can express r_π as

$$r_\pi = \frac{\beta}{\beta I_{BQ}/V_T} = \frac{V_T}{I_{BQ}}$$

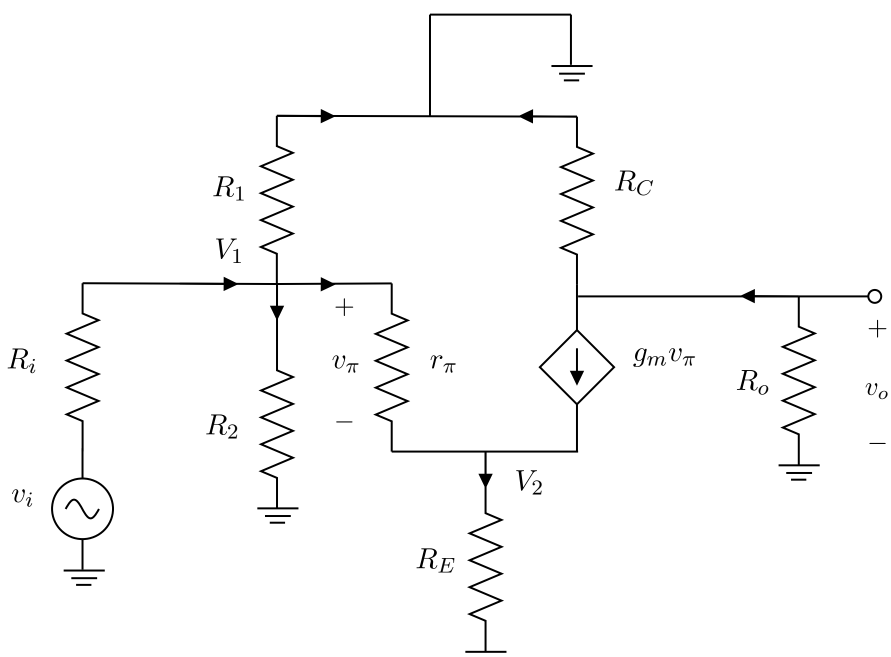
In summary,

$$r_\pi = \frac{V_T}{I_{BQ}}, \qquad g_m = \frac{\beta}{r_\pi}$$

Once again, I_{BQ} and I_{CQ} refer to the quiescent base and collector currents of the BJT. Based on the information from the question, we can calculate

$$r_\pi = \frac{0.025}{1.939 \times 10^{-5}} \approx 1289.23\,\Omega$$
$$g_m = \frac{100}{1289.23} \approx 7.757 \times 10^{-2}\,\mathrm{A/V}$$

Hence, the circuit diagram can be redrawn as



According to Kirchhoff's Current Law,

$$\frac{v_i - V_1}{R_i} = \frac{V_1 - 0}{R_1} + \frac{V_1 - 0}{R_2} + \frac{V_1 - V_3}{r_\pi} \tag{9}$$

$$\Rightarrow \frac{v_i - V_1}{50} = \frac{V_1 - 0}{6000} + \frac{V_1 - 0}{4000} + \frac{V_1 - V_2}{r_\pi}$$

$$\frac{V_2 - 0}{R_E} = \frac{V_1 - V_2}{r_\pi} + g_m v_\pi \tag{10}$$

$$\Rightarrow \frac{V_2}{640} = \frac{V_1 - V_2}{1289.23} + 7.757 \times 10^{-2} v_\pi$$

$$\frac{0 - v_o}{R_o} = \frac{v_o - 0}{R_C} + g_m v_\pi \tag{11}$$

$$\Rightarrow \frac{-v_o}{110} = \frac{v_o}{800} + 7.757 \times 10^{-2} v_\pi$$

$$v_\pi = v_1 - v_2 \tag{12}$$

Solving this system of equations, we yield

$$A_v = \frac{v_o}{v_i} \approx -0.1436 \text{ V/V}$$

By superposition, we obtain

$$\begin{aligned} v_o(t) &= V_C^{DC} + v_C^{AC}(t) \\ &= V_C^{DC} + A_v v_i(t) \\ &= 3.449 - 1.436 \sin(2t) \text{ V} \end{aligned}$$

IV. Conclusion

Although electrical engineers often prefer the Laplace Transform for analyzing linear RLC circuits because it provides a more direct solution, the Superposition Theorem remains an important analytical tool to solve non-linear circuits.

At first glance, this may seem surprising since superposition is strictly valid only for linear systems, whereas devices such as diodes, BJTs, and MOSFETs are inherently nonlinear. The key idea is linearization. By first determining the quiescent (DC operating) point of a nonlinear device, we can approximate its behavior near that operating point with a linear small-signal model. Once the circuit has been linearized, the principles of linear circuit analysis, including superposition, become applicable. This allows engineers to separate the DC bias conditions from the AC signal variations and analyze their effects independently.

As a result, superposition provides both physical insight and computational simplicity, making it a fundamental tool in the design and analysis of amplifiers, analog integrated circuits, communication systems, and many other electronic applications.